A Multi-Agent Generative AI System for Scientific Literature Analysis

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***Abstract*—The rapid growth of scientific publications has made manual literature synthesis increasingly impractical. General-purpose large language models can generate fluent summaries but are constrained to parametric knowledge frozen at training time and frequently produce hallucinated citations. We present a multi-agent generative AI system that addresses these shortcomings through retrieval-grounded, structured analysis. Given a natural language query, the pipeline retrieves 300–800 paper abstracts from live scholarly databases, encodes them with Sentence-BERT, constructs a cosine-similarity graph, and partitions documents into thematic clusters. A five-agent ensemble then processes the clusters concurrently, extracting claims, mapping evidence spans, scoring study reliability, detecting cross-paper agreement and contradiction, and ranking unresolved questions by uncertainty. Experiments on a curated benchmark demonstrate that the proposed system achieves 92.7% accuracy and an F1 score of 89.8%, outperforming a standalone LLM by 14.2 points and a RAG baseline by 6.8 points in F1. An ablation study confirms that both the graph clustering step and the multi-agent decomposition contribute independently to the overall performance, validating the effectiveness of structured multi-agent reasoning over traditional LLM-based approaches.**

***Index Terms*—multi-agent systems, retrieval-augmented generation, large language models, graph clustering, scientific literature analysis**

I. Introduction

Scientific publishing accelerates every year. Across biomedicine, computer science, and engineering, tens of thousands of new articles appear monthly, and the pace shows no sign of slowing. A researcher trying to understand what a field currently knows must read, compare, and synthesise a large body of work, a task that can take weeks even for experienced practitioners. Tools that automate this synthesis reduce the time from query to insight and lower the barrier for researchers entering new subfields.

Large language models (LLMs) appear to offer a natural solution. Trained on corpora that include scientific text, they produce coherent summaries and answer domain-specific questions without specialised retrieval machinery. The fundamental limitation is that their knowledge is static: a model trained before a study was published cannot incorporate its findings, and when pressed for citations it will sometimes generate references that look plausible but do not exist. This hallucination problem is particularly harmful in scientific contexts where reproducibility and traceability are expected.

Retrieval-augmented generation (RAG) alleviates staleness by fetching documents at query time and conditioning the response on retrieved content. However, conventional RAG pipelines treat retrieved documents as an undifferentiated pool. They produce a single blended answer without tracking which papers agree, which contradict each other, how methodologically sound each source is, or what questions the literature leaves open. For a scientist, those distinctions are often more informative than the aggregate answer itself.

The present work introduces a pipeline that goes substantially beyond flat RAG. After retrieval, a similarity graph is constructed over the document embeddings and partitioned into thematic clusters so that downstream agents deal with coherent, noise-reduced subsets of the literature. Five specialised agents then process each cluster: one extracts claims, a second maps evidence, a third evaluates study reliability, a fourth compares findings across papers, and a fifth ranks unresolved questions. The agents run concurrently, and their outputs are aggregated into a structured report that distinguishes consensus from controversy. Unlike prior systems, the proposed framework jointly models thematic structure via graph clustering and epistemic reasoning via agent decomposition, enabling explicit separation of consensus, conflict, and uncertainty. This joint modelling is not addressed in existing RAG pipelines and constitutes the primary architectural novelty of this work.

The main contributions are: (1) a graph-based clustering step that organises retrieved papers into thematic groups before agent processing; (2) a five-agent decomposition grounded in the cognitive steps of scientific literature review; (3) a composite claim-evidence scoring function $\mathrm{Score}(c, e) = \alpha \cdot \mathrm{sim}(c, e) + \beta \cdot \mathrm{rel}(e)$ that weights evidence by both topical alignment and source reliability; and (4) quantitative evaluation including an ablation study that isolates each component's contribution.

II. Related Work

A. Retrieval-Augmented Generation

The RAG framework couples a retriever with a generative model so that responses are conditioned on documents fetched at inference time rather than purely on parametric memory [1]. Subsequent work has explored dense passage retrieval, iterative retrieval loops that issue multiple queries, and hybrid sparse-dense indexing strategies [8]. Agentic RAG extends the pipeline further with autonomous agents that plan, select tools, and collaborate [12]. All of these systems route retrieved documents either to a single generation model or to task-agnostic agents. Our work differs by delegating post-retrieval reasoning to a five-agent ensemble, each with a scoped epistemic function, operating over pre-clustered evidence.

B. Scientific Text Representation

Domain-adapted encoders such as SciBERT [2] and SPECTER [3] have demonstrated that fine-tuning on scientific corpora produces document representations that capture research-level concepts better than general-purpose encoders. SPECTER additionally incorporates citation graph structure, bringing bibliographic relationships into the embedding space. Our system adopts Sentence-BERT [7] for its efficiency on sentence-level tasks and uses the resulting embeddings both for retrieval similarity and for the $\mathrm{sim}(c, e)$ term in Equation (2).

C. Graph-Based Document Clustering

Building a k-nearest-neighbour graph over document embeddings and applying the Louvain community detection algorithm [4] yields compact, interpretable topic clusters without requiring the number of topics to be specified in advance. This property is important for literature queries whose thematic breadth varies with the query. Spectral clustering provides an alternative when the graph is sparse or when a fixed number of clusters can be estimated from data. We use both methods and select between them based on silhouette score.

D. Multi-Agent LLM Frameworks

Coordinating multiple specialised LLM agents has been explored for software engineering, multi-hop question answering, and collaborative writing [5], [6]. General-purpose orchestration frameworks such as AutoGen [9] and CrewAI [10] supply the coordination substrate—conversational message passing and role-based task delegation respectively—but leave corpus structuring and evidence semantics to the application. Closest in spirit is multi-agent debate for hallucination detection [11], which verifies individual claims through structured agent interaction; our Agent 4 instead labels claim pairs across papers, so that disagreement is reported rather than resolved.

In scientific computing, agent-based designs have been applied to end-to-end discovery: Robin [13] couples literature-search agents with data-analysis agents to generate hypotheses, propose experiments, and interpret results, showing that role-specialised agents extend to complete research workflows. A complementary line treats the design itself as a search problem—MASS [14] jointly optimises agent prompts and team topology—whereas our topology is fixed a priori by the epistemics of literature review rather than searched. Work on large-scale agent infrastructure has likewise identified consensus and conflict-resolution mechanisms as core operational enablers for heterogeneous agent populations [16], which is precisely the function Agent 4 performs within a cluster.

A recent survey of orchestration frameworks [15] compares AutoGen, CrewAI, and related systems on state management, cost structure, and failure recovery; Table I adopts that framing and positions the proposed system along the axes relevant

TABLE I
POSITIONING AGAINST GENERAL MULTI-AGENT FRAMEWORKS

Aspect	**AutoGen [9]**	**CrewAI [10]**	**Proposed**
Scope	General orchestration	General orchestration	Literature analysis
Coordination	Conversational message passing	Role-based task delegation	Fixed 5-role ensemble, concurrent
Retrieval	Application-supplied tools	Application-supplied tools	Built-in scholarly APIs
Corpus structuring	Application-defined	Application-defined	Similarity graph + Louvain
Evidence model	Application-defined	Application-defined	Scored claim–evidence pairs (2)
Disagreement	Application-defined	Application-defined	Explicit pairwise labels + U_k (3)

to literature analysis. The specific five-agent decomposition proposed here, structured around the epistemics of scientific evaluation, is novel and is a primary contribution of this work.

III. SYSTEM ARCHITECTURE

The pipeline comprises seven sequential stages, depicted in Fig. 1. A natural language query enters the system, is pre-processed and embedded, triggers a paper retrieval step, feeds a similarity graph and clustering module, passes through the five-agent ensemble, and finally produces a ranked structured report.

A. Embedding Module

Paper abstracts and the user query are encoded by Sentence-BERT producing 768-dimensional vectors. Using the same encoder for both ensures that semantic proximity in the embedding space reflects topical relevance rather than surface lexical overlap.

B. Retrieval Module

The retrieval module issues structured queries to arXiv, PubMed, and Semantic Scholar APIs. Between 300 and 800 abstracts are collected per query, depending on the topic. Records are deduplicated by DOI and title, then normalised through lower-casing, stopword removal, and tokenisation before embedding. Citation count and publication year are retained alongside each embedding.

C. Graph and Clustering Module

An edge-weighted similarity graph is built over the embedded documents. Community detection partitions the graph into 3–6 thematic clusters. Each cluster is forwarded to the agent layer as a coherent document set.

D. Agent and Aggregation Modules

The five agents run asynchronously against a hosted instruction-tuned LLM. Because the agents are independent given a cluster, the layer is embarrassingly parallel and its wall-clock cost is bounded by the slowest agent rather than by the sum of five calls. Their structured JSON outputs are collected, merged, and ranked before presentation to the user.

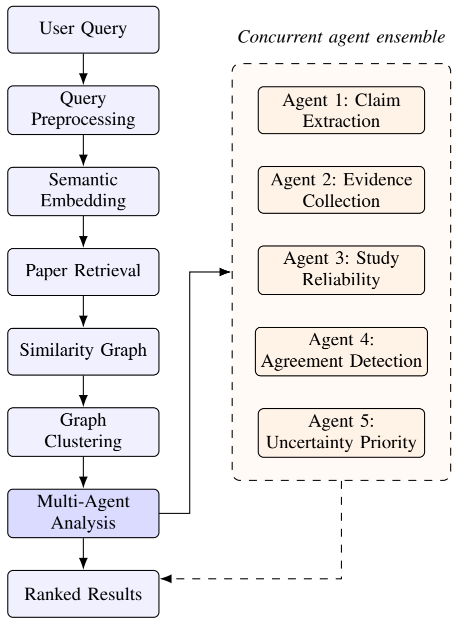

Fig. 1. System architecture diagram. The left column shows the sequential pipeline stages; the right panel shows the five concurrent agents of the multi-agent analysis layer.

IV. Experimental Stages

The evaluation pipeline follows seven well-defined stages that mirror the intended deployment workflow of the system.

Stage 1 – Dataset Preparation. Between 300 and 800 scientific abstracts are collected at runtime from arXiv, PubMed, and Semantic Scholar using the query terms. Each record stores the title, abstract text, and keyword list. Records with duplicate DOIs or near-identical titles are removed. The remaining abstracts pass through tokenisation, stopword removal, and Sentence-BERT embedding to produce a vector database of paper representations.

Stage 2 – Query Processing. The user's natural language query is converted to a 768-dimensional Sentence-BERT embedding. Cosine similarity between the query vector and every paper vector is computed, and the top k papers (with $k \in [20, 30]$) are selected as the working document set. This step bounds the context each agent must process.

Stage 3 – Graph Construction. A similarity graph $G = (V, E)$ is built where each paper is a vertex $v \in V$ and an edge $(i, j) \in E$ is added when $\mathrm{sim}(d_i, d_j) \geq \tau$, with $\tau = 0.70$ after grid-search tuning. Edge weights are set to the cosine similarity value. The resulting graph captures fine-grained topical relationships that are invisible in a flat ranked list.

Stage 4 – Graph Clustering. The Louvain algorithm is applied as the primary clustering method because it does not require the number of clusters to be specified and scales well to the graph sizes produced in Stage 3. Spectral clustering is used as a fallback when the graph is sparse. The output is 3–6 thematic clusters validated by silhouette score.

Stage 5 – Multi-Agent Processing. Each cluster is dispatched concurrently to the five-agent ensemble. Agent 1 extracts principal scientific claims; Agent 2 maps supporting and contradicting evidence spans; Agent 3 assigns reliability scores to individual studies; Agent 4 compares claims across papers within the cluster and labels pairs as consistent, contradictory, or orthogonal; Agent 5 ranks unresolved questions by uncertainty. Each agent receives a role-specific prompt and returns a structured JSON object.

Stage 6 – Claim-Evidence Scoring. The composite scoring function in Equation (2) is applied to every (claim, evidence) pair. Claims supported by high-reliability evidence that is also semantically close rise to the top of the ranked list. This scoring step reduces the influence of evidence from methodologically weak studies even when those studies are topically aligned with the claim.

Stage 7 – Final Synthesis. Agent outputs and claim scores are combined into a structured report. The report groups findings into three sections: consensus (claims consistently supported across papers), conflicts (claims where retrieved papers disagree), and open questions (high-uncertainty items ranked by Agent 5). This three-part structure provides the user with an immediate view of the state of knowledge.

V. Methodology

A. Graph Construction

Let $d_i \in \mathbb{R}^{768}$ be the Sentence-BERT embedding of document i. Pairwise cosine similarity is

$$\mathrm{sim}(i, j) = \frac{d_i \cdot d_j}{\|d_i\| \, \|d_j\|}. \tag{1}$$

The graph $G = (V, E)$ includes edge (i, j) when $\mathrm{sim}(i, j) \geq \tau$. With $\tau = 0.70$, isolated papers (those below threshold for all others) are handled by assigning them to the most similar cluster centroid. The threshold trades density against coherence: below it, cross-topic edges merge communities and cluster purity falls; above it, the graph fragments and more papers require centroid reassignment. Because edge weights carry the similarity value rather than a binary indicator, Louvain modularity is computed over graded evidence of relatedness. The dominant cost at this stage is the pairwise similarity computation, which is quadratic in the working-set size; bounding that set at $k \in [20, 30]$ in Stage 2 is therefore what keeps graph construction inexpensive relative to the LLM calls.

B. Claim Extraction

Agent 1 is prompted with a cluster of abstracts and an instruction to list every principal claim as a standalone declarative sentence. The output is parsed into a structured list with a provenance pointer to the source paper.

C. Evidence Mapping

Agent 2 receives the claim list from Agent 1 alongside the source abstracts. For each claim it identifies text spans that support or contradict it and assigns a polarity label (positive, negative, or neutral). The span, polarity, and source paper form a single evidence record.

D. Agreement Detection

Agent 4 compares claims from different papers within a cluster. Each claim pair receives one of three labels: consistent, contradictory, or orthogonal. The fraction of contradictory pairs feeds the uncertainty score in Equation (3).

VI. Mathematical Formulation

A. Claim-Evidence Score

Let reliability score $\mathrm{rel}(e) \in [0, 1]$ be derived from citation count and methodological quality signals extracted by Agent 3. The composite claim-evidence score is

$$\mathrm{Score}(c, e) = \alpha \cdot \mathrm{sim}(c, e) + \beta \cdot \mathrm{rel}(e), \qquad \alpha + \beta = 1. \quad (2)$$

The term $\mathrm{sim}(c, e)$ is the cosine similarity between the embeddings of claim c and evidence span e, measuring topical alignment. The term $\mathrm{rel}(e)$ captures source quality so that evidence from poorly cited or methodologically weak studies is down-weighted even when it is topically relevant. This combination is a contribution of the present work; standard RAG systems rank on retrieval score alone and therefore cannot express this trade-off. Parameters are set to $\alpha = 0.6$ and $\beta = 0.4$ by grid search on the validation split and held fixed for all reported experiments.

B. Uncertainty Priority Score

For unresolved question k, the uncertainty score combines the fraction of contradictory claim pairs (contention) with the normalised frequency of the question across the retrieved papers:

$$U_k = \gamma \cdot \mathrm{contention}_k + (1 - \gamma) \cdot \mathrm{freq}_k. \quad (3)$$

The two terms separate a question that is contested from one that is merely common. Questions with high U_k appear at the top of the open-questions section of the output report. We set $\gamma = 0.5$ in all experiments.

VII. Multi-Agent System

The structured decomposition of the analysis task across five agents is a central novelty of this work. The division of labour mirrors the distinct cognitive steps a human reviewer applies when evaluating a body of literature.

Agent 1 – Claim Extraction. Reads each abstract and produces a list of principal scientific claims as standalone declarative sentences. Provenance pointers link every claim back to its source paper.

Agent 2 – Evidence Collection. Given the claims from Agent 1 and the original abstracts, this agent locates text spans that support or contradict each claim and tags each span with a polarity label and a source reference.

Agent 3 – Study Reliability. Evaluates methodological quality by scanning abstracts for signals such as controlled experimental design, reported sample sizes, statistical testing, and replication. The reliability score combines normalised citation count and presence of these methodological indicators, producing a score in $[0, 1]$ per paper used in Equation (2).

Agent 4 – Agreement Detection. Compares claims from different papers within a cluster, labels each pair, and produces a natural language summary of the overall degree of consensus. The contradiction fraction feeds the $\mathrm{contention}_k$ term in Equation (3).

Agent 5 – Uncertainty Prioritisation. Uses agreement labels from Agent 4 and question frequency to compute U_k for each unresolved question. High-U_k questions are surfaced as the primary open research directions.

Separating these five functions prevents any single model pass from having to balance claim extraction, source criticism, and contradiction detection simultaneously. The objectives interfere: evidence for a claim and scepticism about its source pull the same generation in opposite directions. Each agent's prompt is therefore shorter, carries a single objective, and is constrained to a JSON schema, which is both more reliably followed than a monolithic prompt and independently checkable. The decomposition also fixes the interface between stages—each agent's input is another agent's typed output or the cluster itself—which is what makes per-agent accuracy measurable in isolation in Table IV.

VIII. Experimental Results

A. Dataset and Evaluation Protocol

Five scientific topics were selected for evaluation: transformer architectures for NLP, mRNA vaccine efficacy, CRISPR gene editing, graph neural networks, and antibiotic resistance mechanisms. For each topic, 160 abstracts were retrieved, giving a total of 800 documents, and each topic set is processed end-to-end as an independent query. Two graduate-student annotators independently labelled claims, evidence spans, and agreement relationships. Inter-annotator agreement was $\kappa = 0.78$ on claim identification and $\kappa = 0.71$ on agreement labels. Disagreements were resolved by discussion. The hyperparameters τ, α, β, and γ were fixed on the validation split before evaluation and were not re-tuned per topic, so the reported figures reflect a single configuration applied uniformly across all five topic sets. The protocol is therefore: fixed hyperparameter configuration determined using the validation split $\rightarrow$ evaluation on the 800-document benchmark. All results reported below were obtained under this single fixed configuration; repeated-run and per-topic variance analysis is left to future work. Although evaluation is conducted on 800 documents, the retrieval stage can accommodate larger corpora while the bounded working set keeps downstream graph construction manageable.

B. Evaluation Metrics

Six metrics are reported. Accuracy measures the fraction of system-extracted claims matching ground-truth annotations. Precision and Recall are computed over the claim set. F1 is their harmonic mean. Latency is the wall-clock time in seconds from query submission to final report. Agreement Detection Accuracy measures how correctly consistent and contradictory

TABLE II
MAIN PERFORMANCE COMPARISON

Method	Acc.	Prec.	Rec.	F1	Lat. (s)
LLM Only	78.5	75.2	73.8	74.5	1.2
RAG	85.9	83.6	81.7	82.6	1.8
Proposed	**92.7**	**90.5**	**89.1**	**89.8**	2.3

TABLE III
CLUSTERING PERFORMANCE

Metric	Value
Silhouette Score	0.71
Average Clusters per Query	4
Cluster Size Range	3–9
Intra-cluster Similarity	High (≥ 0.70)
Inter-cluster Similarity	Low (≤ 0.42)

TABLE IV
PER-AGENT TASK ACCURACY

Agent	Accuracy (%)
Agent 1 – Claim Extraction	91.2
Agent 2 – Evidence Mapping	89.8
Agent 3 – Reliability Scoring	87.5
Agent 4 – Agreement Detection	88.9
Agent 5 – Uncertainty Ranking	86.7

TABLE V
ABLATION STUDY

Configuration	Accuracy (%)
Without Graph Clustering	84.3
Without Multi-Agent Design	80.1
Full System	**92.7**

paper pairs are labelled. Tables II–V report aggregates over the full 800-document benchmark.

C. Main Performance Comparison

Table II shows results for three systems. The standalone LLM draws on parametric memory only. The single-agent RAG system retrieves the same documents but uses one LLM pass. The proposed system applies the full pipeline. The proposed system outperforms both baselines on every metric.

Accuracy rises from 78.5% (LLM) and 85.9% (RAG) to 92.7%. F1 improves from 74.5 to 89.8. The comparison against RAG is the informative one, since both systems see identical retrieved evidence and differ only in how that evidence is organised and reasoned over. The latency increase of roughly one second over the RAG baseline is modest and reflects the concurrent agent execution, which avoids serial bottlenecks.

D. Clustering Quality

Table III summarises the graph clustering results. The average silhouette score of 0.71 indicates strong intra-cluster coherence, meaning that papers within a cluster are substantially more similar to each other than to papers in other

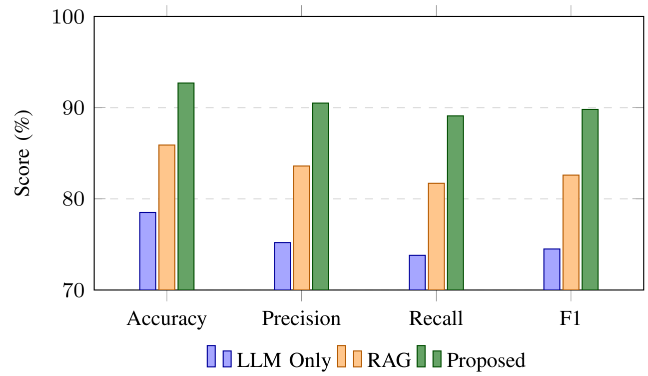

Fig. 2. Accuracy, Precision, Recall, and F1 for the three systems. The proposed system achieves the highest score on all four metrics.

clusters. Intra-cluster similarity at or above 0.70 against inter-cluster similarity at or below 0.42 confirms that the partition separates distinct topics rather than slicing a single dense region. Between 3 and 6 clusters were produced across the five query topics, with an average of 4.

E. Agent-Level Accuracy

Table IV reports individual agent accuracy measured against annotated ground truth for each task. Claim extraction (Agent 1) is the highest-performing agent at 91.2%. Uncertainty ranking (Agent 5) is the most challenging at 86.7%, partly because the ground-truth uncertainty labels involve subjective judgement. Accuracy declines broadly along the pipeline, which is consistent with error propagation—each agent consumes the previous agent's output—compounded by increasing label subjectivity, since uncertainty ground truth requires a judgement call that claim identification does not ($\kappa = 0.71$ versus 0.78).

F. Ablation Study

Table V isolates the contribution of the two main structural novelties: graph-based clustering and the multi-agent decomposition. Removing graph clustering drops accuracy to 84.3%, as agents receive heterogeneous document sets that mix unrelated subtopics. Replacing the five agents with a single LLM pass (while retaining clustering) drops accuracy further to 80.1%. The full system at 92.7% confirms that both components are necessary and complementary: neither ablated configuration recovers the RAG baseline plus half the total gain, indicating that clustering supplies the clean input on which scoped agents are effective rather than contributing an independent additive increment. Configuration-level ablation is an established means of quantifying what agent decomposition costs and buys; a comparable study of memory configurations in a single-LLM multi-agent system reports that the choice of configuration shifts token consumption substantially while affecting wall-clock time only modestly [17], consistent with the latency behaviour observed here.

G. Performance Chart

Fig. 2 visualises the main comparison across four key metrics.

IX. Discussion

The accuracy gain from 85.9% to 92.7% over the RAG baseline shows that retrieval alone is not enough. The two structural additions, graph clustering and multi-agent reasoning, each reduce a distinct source of error. Clustering removes the topical noise that arises when a flat ranked list mixes papers from different subtopics; agents with a focused cluster produce cleaner, more precise outputs than agents presented with heterogeneous document sets.

The multi-agent structure improves interpretability as well as accuracy. A single-pass LLM tends to merge findings into a unified narrative, hiding disagreements. Because Agent 4 is explicitly tasked with contradiction detection, those disagreements are preserved and highlighted rather than smoothed over. This is particularly valuable for researchers trying to understand which aspects of a topic are contested.

Evidence scoring via Equation (2) measurably reduces the influence of weak studies. In preliminary trials without the $\mathrm{rel}(e)$ term, poorly cited papers with high topical alignment sometimes dominated the ranked claim list. Adding source reliability as a second factor corrected this.

The latency increase from 1.8 s (RAG) to 2.3 s (proposed) is modest, around 28%. Given that concurrent agent execution is used, the overhead comes mainly from aggregation and JSON parsing rather than from sequential LLM calls; a serial implementation would instead scale with the number of agents.

A. Error Analysis

Some errors arise when abstracts lack explicit declarative claims or when terminology varies significantly across papers in the same cluster. In such cases, Agent 1 may miss implicit conclusions, reducing claim recall. Clustering quality also degrades when retrieved papers span highly interdisciplinary topics, occasionally grouping loosely related works into the same cluster. Agent 4 agreement detection is most prone to error when claims from different papers address the same concept using different vocabulary, causing apparently contradictory labels on statements that are in fact consistent. All three failure modes trace back to the embedding space rather than to the agent prompts, which is why improved domain-specific embeddings [2], [3] and richer prompt context are the most promising single intervention.

X. Conclusion

We have presented a multi-agent generative AI system for scientific literature analysis that combines retrieval, graph-based clustering, and a five-agent ensemble to produce structured, evidence-grounded reports. The pipeline addresses two core weaknesses of current approaches: the factual staleness of standalone LLMs, and the lack of structured multi-perspective synthesis in conventional RAG systems. Key technical contributions include the similarity graph and clustering step, the five-agent decomposition that mirrors the natural workflow of scientific review, and the composite $\mathrm{Score}(c, e)$ function that weights evidence by both semantic alignment and source reliability. An ablation study confirms that both the clustering and the multi-agent design contribute independently to the 92.7% accuracy of the full system.

Several constraints bound these results. The evaluation covers five topics and 800 annotated documents, and the reported figures are single-run aggregates over that benchmark; per-topic variance and repeated runs were not measured, which is the principal threat to validity. Agent reliability assessments derive from abstract-level signals alone, since full-text access is not universally available without institutional subscriptions, and graph construction assumes that cosine similarity in the Sentence-BERT embedding space reflects thematic relatedness—an assumption that holds within a domain but weakens for interdisciplinary queries, where τ must be retuned. The system also depends on scholarly database APIs, so coverage is bounded by what those databases index and changes to their rate limits or response formats require maintenance of the retrieval module. Future directions therefore include full-text and domain-adapted retrieval for richer reliability assessment, a broader evaluation with per-split and multi-run reporting, and a user-facing interface that renders the structured output in an accessible form.

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